

CQE Paper 32 - The Supervisor Cursor: Superpermutations as the Compressed Dimensional Action Graph

CQE-paper-32

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-12

Construction Basis

Use superpermutation schedules as supervisor cursors: compressed action graphs that enumerate every requested ordering while producing deployable PermForge/GraphStax machinery.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- production paper body pending rewrite: production\papers\CQE-paper-32\PAPER-BODY.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-32\paper_kernel_manifest.json
- body: production\papers\CQE-paper-32\PAPER-BODY.md
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- formal: production\papers\CQE-paper-32\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-32\02-CQE-tool\TOOL.md
- workbook: production\papers\CQE-paper-32\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-32.md

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Proof/Exposure Hierarchy

The proof-carrying content of this paper is the mathematics: the definitions, lemmas, constructions, examples, and receipts that establish the claimed transport. Paper 00, workbook sheets, analog tools, and open-obligation ledgers are supplemental validation and exposure layers. They exist to make the math inspectable, reproducible, and accessible without requiring a particular software stack. In the simplest case, the same state transitions can be marked with ordinary physical tokens, lines, or dirt; the point is not the material, but the preserved center, boundary, transform, residue, and receipt structure.

****Series note.**** Papers 00-31 built the substrate. Paper 32 opens the applied series: the methods of the corpus pointed at problems that look arbitrary from outside - and the demonstration that solving them produces working machinery (here: PermForge, the supervisor-cursor engine of ChromaBlend Studio) while the problem's own structure lands, term for term, inside the corpus mathematics.

1. The seemingly arbitrary application

The superpermutation problem asks: what is the shortest string over n symbols containing every permutation of those symbols as a contiguous substring? It reads as a curiosity of recreational combinatorics. It is not. Within this corpus it is the exact statement of a question the substrate already forces:

****If C only exists when an enumeration is requested ($\text{Gamma}(s) = _C(\text{enum}(r_i, W))$, Paper 15's emission law read operationally), then what is the complete, minimal schedule of all enumeration requests at scale n ?****

That schedule is the superpermutation. The system reads dimensions via tape in slots; an ordering of n slot-reads is a permutation; the set of all orderings is the dimensional action graph at $D = n$ ($n!$ vertices); and the superpermutation is that graph serialized onto a 1D tape with maximal overlap sharing - its ****full compression****:

$\text{superperm}(N) = \text{compress}(\text{ActionGraph}(D = N))$

Live numbers: 24 readings in 33 symbols ($n=4$, $2.91\times$ over naive), 120 in 153 ($n=5$, $3.92\times$), 40320 in 46205 ($n=8$, $6.98\times$). The cursor string is a supervisor, never content: walking it fires every possible ordering of slot-reads exactly once.

2. Walking the power-of-ten ladder: solving each $N+1$

The classic recursive construction is the chart walk made literal. To lift scale n to $n+1$: visit every permutation of scale n ****in the order the cursor visits them****, and thread the new symbol through each one (block $p \cdot (n+1) \cdot p$, merged with maximal overlap). Every local chart of scale n becomes a sheet of the $(n+1)$ -cursor - local at resolution R is global at resolution $R-1$, the MDHG ladder walked one rung per symbol.

Executed live from the single character "1" (total walk time 0.04 s):

| n | length | $\log_{10}$ | $= k!$ | coverage | MDHG rung |

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|---|-----|-----|-----|-----|-----| 1 | 1 | 0.00 | yes | yes | grain | | 2 |
3 | 0.48 | yes | yes | dust | | 3 | 9 | 0.95 | yes | yes | triad | | 4 | 33 | 1.52 | yes | yes |
block | | 5 | 153 | 2.18 | yes | yes | cluster | | 6 | 873 | 2.94 | yes | yes | domain | | 7 |
5913 | 3.77 | yes | yes | region | | 8 | 46233 | 4.67 | yes | yes | planet |

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Each N+1 solve is one rung up the power-of-ten scale - $\log_{10}$ advances by $\sim \log_{10}(n)$ per rung, the factorial walk. The ninth rung (universe) is the asymptote the ladder names but does not reach: there is no terminal N.

3. All complexity effects are inside the local charts

Everything difficult about *minimality* is a property of the length-n windows - the local charts - not of the global string:

- ****n=4 (block):**** the chart structure admits exactly ONE minimal cursor, and it is palindromic - equal to its own LR-podal reversal. 24 permutations = the 24 roots of D4 = the vertices of the 24-cell, the unique self-dual regular 4-polytope. A 4D object with a self-mirrored schedule: no torsion.
- ****n=5 (cluster):**** minimality settled by exhaustive search (Chaffin, 2014): exactly ****8 minimal cursors**** of length 153 - one palindrome and seven symmetry-broken trees. ****8 solutions = 8 dimensions = the E8 lane count.**** The 4D object's schedule, lifted one symbol, demands the full 8D ambient space. The seven trees admit no canonical choice: under relabeling $\times$ reversal the octad is a homogeneous space - a torsor. Choosing a cursor is choosing a gauge. This is the GR torsor effect of the dimensional lift.
- ****The wrap correspondence (recorded observation, pending validation):**** the reversal involution acts on the octad with 4 fixed points + 2 swapped pairs ($\{0,1,4,6\}$ fixed; $2 \leftrightarrow 5$, $3 \leftrightarrow 7$) - the same orbit type as the 8 chart states under swap_LR (4 Lie conjugates fixed + 2 chiral pairs), the involution whose invariant defines the gluon $\Gamma(s) = C$. The schedule space at n=5 and the state space at n=3 wrap identically.
- ****n>=6 (domain and up):**** the charts stop admitting closed answers. Minimality is open; only bounds and search survive. The chart walk gives k!; SAT search broke it at n=6 ($872 < 873$, Houston 2014); the question becomes a corridor between a proven floor and a constructed ceiling.

4. The n=8 attempt - standing on Egan, Houston, Johnston, Chaffin

Nothing here is reinvented. The lower bound is the Houston-Pantone-Vatter formalization (2018) of the 2011 anonymous bound:

$$> L(n) \geq n! + (n-1)! + (n-2)! + n - 3$$

The ceiling is Egan's construction (2018, building directly on Houston's minimality break):

$$> L(n) \leq n! + (n-1)! + (n-2)! + (n-3)! + n - 3$$

Their difference is exact and structural: ****Egan - floor = (n-3)!**** - the open corridor at every scale is one factorial wide, three rungs down the ladder.

The n=8 attempt, executed live (0.06 s):

- ****Chart-walk construction from scratch:**** 46233 symbols (= k!), full coverage of all 40320 permutations of 8 symbols - verified.
- ****The shipped field record**** (curated per Johnston's records): 46205 symbols, full coverage - verified.

- ****46205 = Egan's formula at n=8 exactly**** ($40320 + 5040 + 720 + 120 + 5$). The best known n=8 cursor IS Egan's construction; it saves precisely 28 symbols over the chart walk.
- ****The open window:**** floor 46085, ceiling 46205, width **** $120 = 5! = (n-3)!$ ****. Search has entered this corridor at smaller scales (n=6: -1 below the formula; n=7: -2, Egan 2019); at n=8 the corridor is untouched.

This is the best solution for n=8 the established methods admit, confirmed inside this format: their constructions and bounds, our verifiers and charts. The $(n-3)!$ corridor is the standing obligation.

5. What the application produced

Solving this "arbitrary" problem inside the corpus produced operational machinery, not commentary:

- ****PermForge**** (``GraphStax.permforge``): embedded n=4/n=5 cursor tables, the octad with its computed reversal orbit, the bounds ladder, the $N \rightarrow N+1$ lift, coverage verifiers, and the live n=8 exhibit - all import-time lookup, no recomputation.
- ****The supervisor scheduler**** (``SuperPermScheduler``): n=4 blocks of services dispatched in cursor order - every ordering of 4 services fired exactly once per 33-step pass (the service-wiring pattern of the donor stack, now a library primitive).
- ****Supervised enumeration**** (``GraphStaxEngine.enumerate_ribbon``): a ribbon's bits resolved into C-gluon Stax identities ****in cursor order**** - the first request per position is the C-production, repeats are idempotent lookups ($f(f(x)) = f(x)$). The C-term's dependence on requested enumeration is now executable, not aspirational.

6. Open obligations (Paper 32 ledger)

- ****O-32.1**** - The n=8 corridor: 120 symbols between 46085 and 46205. Egan's n=7 search methods (kernel graphs over 2-cycle/3-cycle quotients) transported to n=8 inside this format.
- ****O-32.2**** - Validate the octad-orbit $\leftrightarrow$ chart-state-orbit correspondence ($4+2+2 = 4+2+2$) as a theorem rather than an observation; identify the functor between schedule space at n and state space at n-2.
- ****O-32.3**** - Ship the search records below the formula at n=6 (872) and n=7 (5906) as field data alongside the construction records, with coverage receipts.
- ****O-32.4**** - The ninth rung: formalize the universe-level asymptote of the ladder ($\lim \log_{10}$ behavior) in the corpus's scale language.

Citation chain

- Chaffin, B. (2014): exhaustive minimality at n=5 - the octad.
- Houston, R. (2014): SAT discovery $872 < 873$ at n=6 - the chart walk is not minimal.
- Houston, R., Pantone, J., Vatter, V. (2018): formalization of the lower bound $n! + (n-1)! + (n-2)! + n - 3$ (anonymous origin, 2011).
- Egan, G. (2018-2019): the upper construction $n! + (n-1)! + (n-2)! + (n-3)! + n - 3$; search records at n=7.
- Johnston, N.: curation of minimal superpermutation records and data.
- This corpus: Paper 15 (emission law / observer chain), Paper 03 (S3 wraps), Paper 05 (oloid

carrier), MDHG ladder, ribbon 8-slot transport.

Compiled from CQECMPLX-Production master corpus. Verifiers executed live 2026-06-09; all components verified complete.